

SIMÓN BARRERA RUÍZ

Mechatronics Engineer · AI Specialist · Robotics Technical Leader

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PROFESSIONAL SUMMARY

Mechatronics Engineering student with a high-impact track record in Artificial Intelligence consulting, neural-networks-driven solutions, and international competitive robotics. Founder of an AI corporate training firm (SCIBARU AI), legal representative and operations lead at a last-mile logistics startup, and international technical mentor for Team Colombia at FIRST Global Challenge — achieving Top 4 Worldwide (2024). Interdisciplinary profile combining embedded systems, mechanical design, generative AI, and strategic leadership, with proven ability to deliver real-world technological impact across educational, business, and institutional contexts.

CORE COMPETENCIES & TECHNICAL SKILLS

- ▶ Artificial Intelligence & Machine Learning
- ▶ Neural Networks & Deep Learning (applied)
- ▶ Generative AI — LLMs, Prompt Engineering
- ▶ Python · R · Java · C (Arduino) · HTML
- ▶ Mechatronics & Embedded Systems
- ▶ Robotics Design & Systems Integration
- ▶ SolidWorks · Onshape · 3D Modeling & Simulation
- ▶ MATLAB · Octave · RStudio
- ▶ Arduino IDE · Proteus · Multisim · TinkerCAD
- ▶ Data Analysis & Operational Dashboards
- ▶ Agentic IDEs: Google Antigravity · Google AI Studio
- ▶ Full-Stack Web Dev · VS Code · Android Studio
- ▶ Industry 4.0 · Digital Transformation Strategy

PROFESSIONAL EXPERIENCE

Founder & Director | SCIBARU AI — Palmira, Colombia

2023 – Present

Company website: scibaru-ai.vercel.app

AI corporate training and consulting firm bridging theory and real-world Industry 4.0 adoption.

- ▶ Designed and delivered 6+ AI training programs across independent entities and institutional clients, reaching diverse populations including senior citizens (60+) and educational professionals.
- ▶ Delivered 2 full courses at Centro de Bienestar Sagezza: 'Introduction to AI' and 'AI Applied to Active Life' — specially designed for adults 60+, demonstrating the ability to communicate complex AI concepts accessibly.
- ▶ Delivered 2 courses at Colegio Bilingüe San José Campestre (administrative staff and teaching faculty) and served the institution as AI Consultant.
- ▶ Delivered courses on agentic AI systems — including platforms such as Google Antigravity and Manus — to Eduqualityco, covering real-world automation.
- ▶ Delivered 2 complete courses on generative AI label design and prompt engineering for traditional Viche and Pacific Coast Colombian cuisine entrepreneurs, in partnership with Eduqualityco.
- ▶ Delivered 3 additional courses for independent organizations, advising on AI adoption and workflow automation.
- ▶ Advised organizations on identifying, structuring, and executing high-ROI AI projects with measurable impact timelines.
- ▶ Positioned SCIBARU AI as a reference for Industry 4.0 consulting in the region, serving clients from the Chamber of Commerce ecosystem.

Operations Coordinator & Legal Representative | BaruCourier SAS — Palmira

2024 – Present

End-to-end operational and strategic leadership of a last-mile logistics startup.

- ▶ Managed traceability and delivery control of 27,500+ packages per month through data systematization and operational dashboards.
- ▶ Developed BaruLogix — a proprietary web platform for logistics tracking, re-dispatch coordination, and data-driven decision-making.
- ▶ Led intercompany coordination, labeling, sorting, and integration with external delivery partners, ensuring operational continuity at scale.

Full-Stack Web Developer | Independent & Corporate Projects

Aug 2024 – Present

Sole developer across 7+ production web projects using AI-assisted development (Google Antigravity, Google AI Studio, vibe coding methodology).

- ▶ Clients include: BaruCourier SAS (BaruLogix platform), Colegio Bilingüe San José Campestre, Eduqualityco, Carnitas by Mr. Cousins (restaurant — Palmira), and additional independent organizations.
- ▶ Delivered end-to-end solutions — from requirements analysis to deployment — across logistics, education, and corporate sectors.
- ▶ Leveraged agentic IDEs (Google Antigravity) and Google AI Studio to accelerate development cycles and ensure production-quality output as a single developer.

International Technical Mentor | Team Colombia: Educational Innovation Foundation · FIRST Global Challenge

2023 – Present

Technical mentorship in FIRST Global Challenge — the largest international FIRST event globally, representing Colombia among 190+ nations.

- ▶ **Top 4 Worldwide (2024) · Top 8 Worldwide (2020):** served as a key contributor in structural optimization, regulatory compliance, and competition strategy.
- ▶ 3rd Place Worldwide — Engineering Documentation (2024): played a key role in the engineering documentation process that earned global recognition.
- ▶ Designed and implemented educational strategies and training programs for high school students in audiovisual production and international relations within the FIRST ecosystem.
- ▶ Led international relations, communication strategy, and social media management, achieving global recognitions for the team (2022–2025).

Technical Mentor — FTC (First Tech Challenge) | Team Colombia: Educational Innovation Foundation 2025 – Present

Co-founded one of the first FTC teams in Colombia, preparing for the country's inaugural FTC Regional in Cartagena (2026).

- ▶ Led mechanical design, robot construction, materials selection, and build strategy aligned with international FIRST standards.

Technical Mentor — FRC (First Robotics Competition) | Yumbo Mayor's Office · Stemachines 9087 Jan – May 2023

Educational robotics project led by the Mayor's Office of Yumbo, executed through Fundación Team Colombia.

- ▶ Designed chassis, mechanisms, and technical documentation in full compliance with FIRST Robotics Competition regulations.

Digitalization Consultant | Eduqualityco · Colombian Ministry of Culture

2024

'Pacífico en su Salsa' project — digitalization and optimization of educational processes.

- ▶ Structured and optimized educational content for digital learning environments, integrating technology, culture, and pedagogical strategy.
- ▶ Implemented technological tools for educational delivery aligned with Ministry of Culture standards.
- ▶ Co-led 2 complete training courses on generative AI and prompt engineering applied to label design for Viche and traditional Pacific Colombian cuisine entrepreneurs — bridging cultural heritage and applied AI.

Academic Monitor & Peer Mentor | Universidad Autónoma de Occidente — COMPAS Program 2024

- ▶ Provided academic and social support for first-semester students during university adaptation, facilitating integration into academic life.
- ▶ Designed and applied formative strategies for civic education, strengthening students' critical thinking, social responsibility, and collaborative skills within an interdisciplinary university context.

Technical & Audiovisual Advisor | Colegio Bilingüe San José Campestre · Eduqualityco · Carnitas by Mr. Cousins 2023 – Present

- ▶ Technical and creative advisory for student projects, institutional events, and business branding initiatives across educational and commercial clients.
- ▶ Produced audiovisual materials and provided strategic communication support for Colegio Bilingüe San José Campestre, Eduqualityco, and Carnitas by Mr. Cousins (Palmira).

Referee & Judge | MakeX Robotics · FIRST LEGO League As applicable

- ▶ Referee at MakeX (Yumbo): evaluated robot performance and enforced competition rules.
- ▶ Judge at FIRST LEGO League (Cali): assessed robot design, FIRST Core Values, and team spirit.

Technical Project Advisor | Educational Institutions As applicable

- ▶ Mentored students in applied engineering: prototype design, embedded systems, electronics fundamentals, and Arduino programming.

COMPETITIVE ACHIEVEMENTS & AWARDS

- Top 4 Worldwide — FIRST Global Challenge, 2024
- 3rd Place Worldwide — Engineering Documentation, 2024
- Top 8 Worldwide — FIRST Global Challenge, 2020
- 1st Place — Ingenium Fest, UAO, 2025
- 2nd Place — Ingeniathon, UAO, 2024 & 2025
- International Social Media Recognition, Team Colombia (2022–2025)
- Academic Excellence Award — Palmira Mayor's Office, Dec 2023
- FIRST Global Challenge Competitor — Team Colombia (2020–2022)
- International Volunteer — FIRST Global Challenge 2025 (Panama City)
- Accepted Volunteer — FIRST Global Challenge 2026 (Seoul, South Korea)

ACADEMIC BACKGROUND

B.Sc. in Mechatronics Engineering (In Progress) Universidad Autónoma de Occidente — Cali, Colombia 2023 – Present	High School Diploma Colegio Bilingüe San José Campestre Graduated 2023
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CERTIFICATIONS & ADDITIONAL TRAINING

- ▶ Generative AI — PALMUS Ventures / Palmira Chamber of Commerce
- ▶ Prompt Engineering — Google
- ▶ Prompt Engineering — Udemy
- ▶ 3D Modeling — Universidad Autónoma de Occidente
- ▶ **Claude Architect — Certified by Anthropic**
- ▶ Innovative Education — Fundación Team Colombia
- ▶ Data Privacy & Protection — FIRST
- ▶ Volunteer Coordination — FIRST
- ▶ Emotional First Aid — Universidad Autónoma de Occidente

▸ Audiovisual Production — FIRST Global

▸ Gracious Professionalism — FIRST

LANGUAGES

Spanish — C2 Native

English — C1+ (Cambridge Linguaskill certified)

German — A1 (Duolingo)

French — A1 (Universidad Autónoma de Occidente)

References available upon request · LinkedIn & portfolio available on request